3. Matplotlib

Introduction

Matplotlib is one of the most popular Python libraries used for data visualization. It helps users create various types of graphs and charts to understand patterns, trends, and relationships in data. Matplotlib supports line graphs, bar charts, scatter plots, histograms, pie charts, box plots, and many other visualization techniques. It provides extensive customization options such as colors, labels, titles, legends, markers, and gridlines. The library integrates seamlessly with NumPy and Pandas, making it an essential tool for Data Science and Machine Learning. Matplotlib enables users to present complex data in a simple, clear, and visually appealing manner.

Import Library

```
import matplotlib.pyplot as plt
```

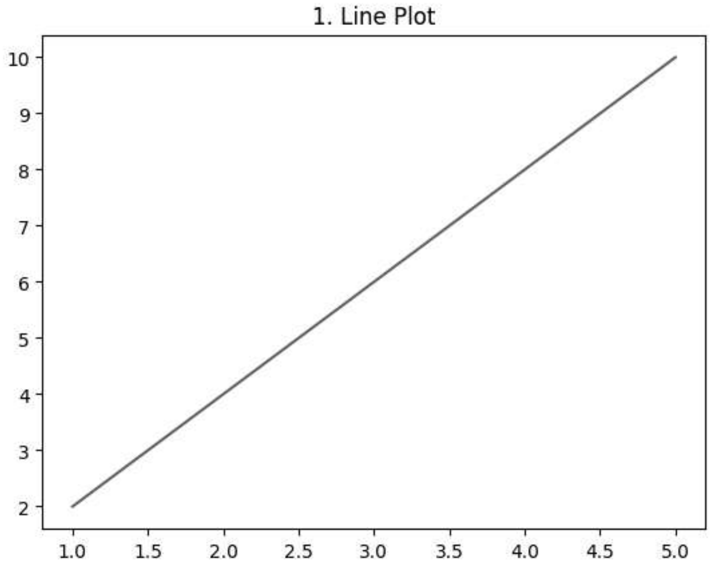
Common Functions

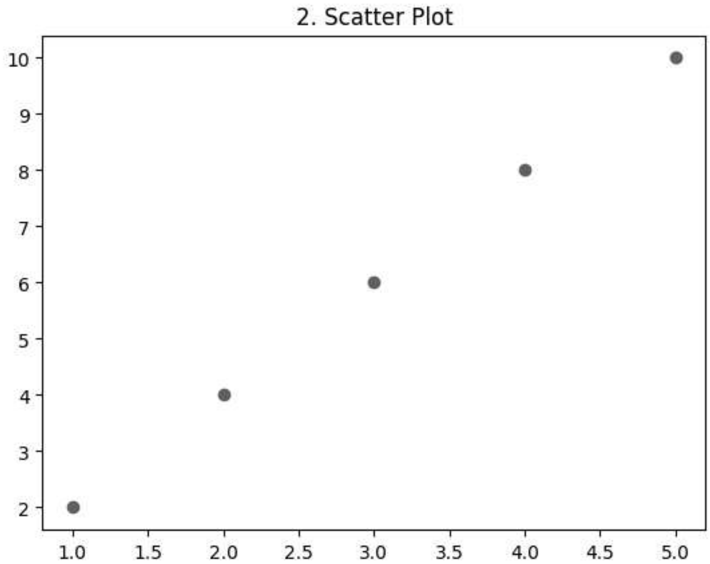
1. **plot()** – Creates a line plot.
2. **scatter()** – Creates a scatter plot.
3. **bar()** – Creates a bar chart.
4. **barh()** – Creates a horizontal bar chart.
5. **hist()** – Creates a histogram.
6. **pie()** – Creates a pie chart.
7. **boxplot()** – Creates a box plot.
8. **xlabel()** – Sets the label of the x-axis.
9. **ylabel()** – Sets the label of the y-axis.
10. **title()** – Adds a title to the graph.
11. **legend()** – Displays the legend.
12. **grid()** – Displays grid lines.
13. **xlim()** – Sets x-axis limits.
14. **ylim()** – Sets y-axis limits.
15. **xticks()** – Sets x-axis tick positions.
16. **yticks()** – Sets y-axis tick positions.
17. **figure()** – Creates a new figure.
18. **subplot()** – Creates multiple plots in one figure.
19. **savefig()** – Saves the figure as an image.
20. **show()** – Displays the graph.
21. **fill_between()** – Fills the area between curves.
22. **stem()** – Creates a stem plot.
23. **step()** – Creates a step plot.
24. **errorbar()** – Displays error bars on data.
25. **text()** – Adds custom text to the graph.

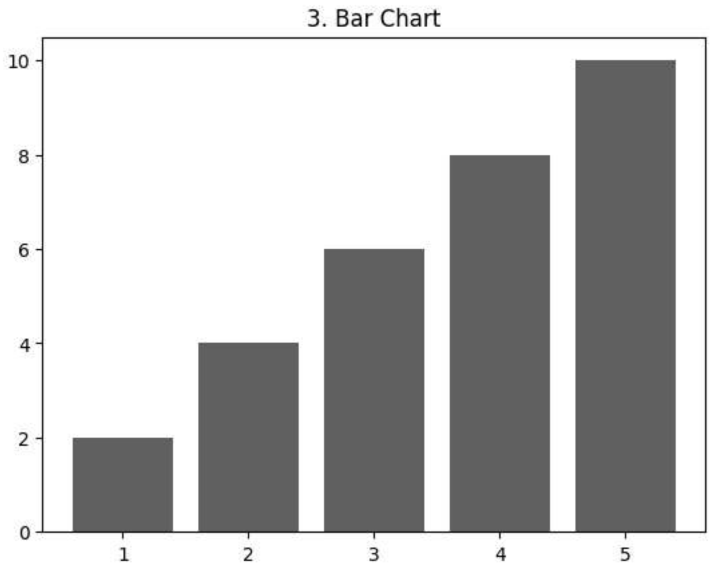
```
import matplotlib.pyplot as plt
import numpy as np

x = np.array([1,2,3,4,5])
y = np.array([2,4,6,8,10])

# 1. plot()
plt.figure()
plt.plot(x, y)
plt.title("1. Line Plot")
plt.show()

# 2. scatter()
plt.figure()
plt.scatter(x, y)
plt.title("2. Scatter Plot")
plt.show()

# 3. bar()
plt.figure()
plt.bar(x, y)
plt.title("3. Bar Chart")
plt.show()
```

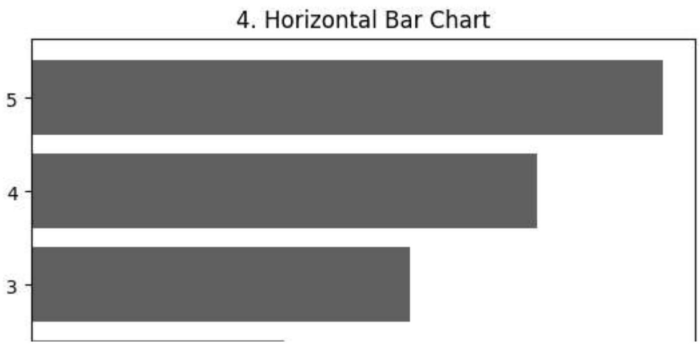
```
# 4. barh()
plt.figure()
plt.barh(x, y)
plt.title("4. Horizontal Bar Chart")
plt.show()

# 5. hist()
plt.figure()
plt.hist([1,2,2,3,3,3,4,4,5])
plt.title("5. Histogram")
plt.show()

# 6. pie()
plt.figure()
plt.pie([30,40,30], labels=["A","B","C"], autopct="%1.1f%%")
plt.title("6. Pie Chart")
plt.show()

# 7. boxplot()
plt.figure()
plt.boxplot([10,20,15,30,25,35])
plt.title("7. Box Plot")
plt.show()

# 8-16. Labels, title, legend, grid, limits and ticks
plt.figure()
plt.plot(x, y, label="Data")
plt.xlabel("X-axis")           # 8
plt.ylabel("Y-axis")           # 9
plt.title("8-16 Functions")    # 10
plt.legend()                   # 11
plt.grid(True)                 # 12
plt.xlim(0,6)                  # 13
plt.ylim(0,12)                 # 14
plt.xticks(x)                  # 15
plt.yticks(range(0,13,2))      # 16
plt.show()

# 17. figure()
plt.figure(figsize=(6,4))
plt.plot(x,y)
plt.title("17. Figure")
plt.show()

# 18. subplot()
plt.figure(figsize=(8,3))
plt.subplot(1,2,1)
plt.plot(x,y)
plt.title("Plot")
plt.subplot(1,2,2)
plt.bar(x,y)
plt.title("Bar")
plt.show()

# 19. savefig()
plt.figure()
plt.plot(x,y)
plt.title("19. Save Figure")
plt.savefig("graph.png")
plt.show()

# 20. show()
plt.figure()
plt.plot(x,y)
plt.title("20. Show Graph")
plt.show()

# 21. fill_between()
plt.figure()
plt.plot(x,y)
plt.fill_between(x,y,color="lightblue")
plt.title("21. Fill Between")
plt.show()

# 22. stem()
plt.figure()
plt.stem(x,y)
plt.title("22. Stem Plot")
plt.show()

# 23. step()
plt.figure()
```

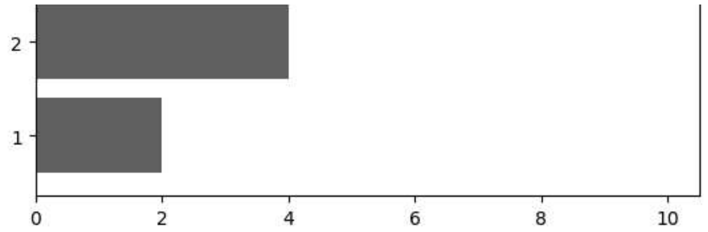
```
plt.step(x,y)
plt.title("23. Step Plot")
plt.show()

# 24. errorbar()
plt.figure()
plt.errorbar(x,y,yerr=0.5,fmt='o')
plt.title("24. Error Bar")
plt.show()

# 25. text()
plt.figure()
plt.plot(x,y)
plt.text(3,6,"Center Point")
plt.title("25. Text")
plt.show()
```

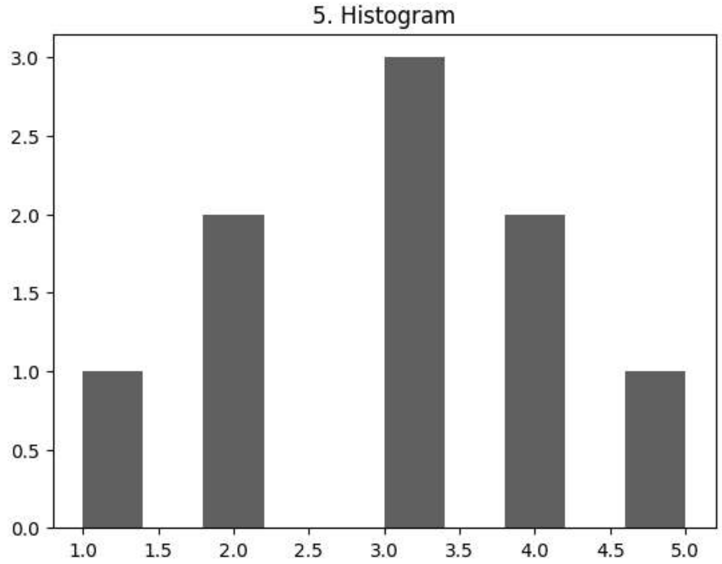

1. Line Plot
10
9
8
7
6
5
4
3
2
1.0
1.5
2.0
2.5
3.0
3.5
4.0
4.5
5.0

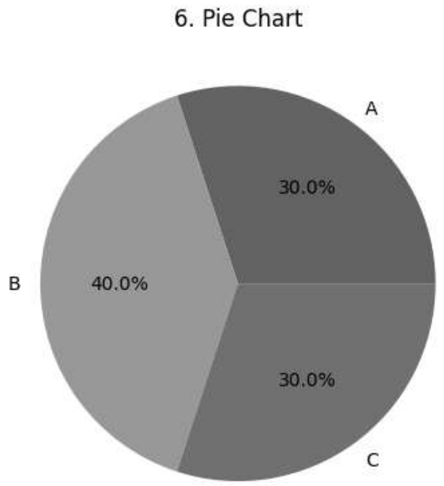

2. Scatter Plot
10
9
8
7
6
5
4
3
2
1.0
1.5
2.0
2.5
3.0
3.5
4.0
4.5
5.0

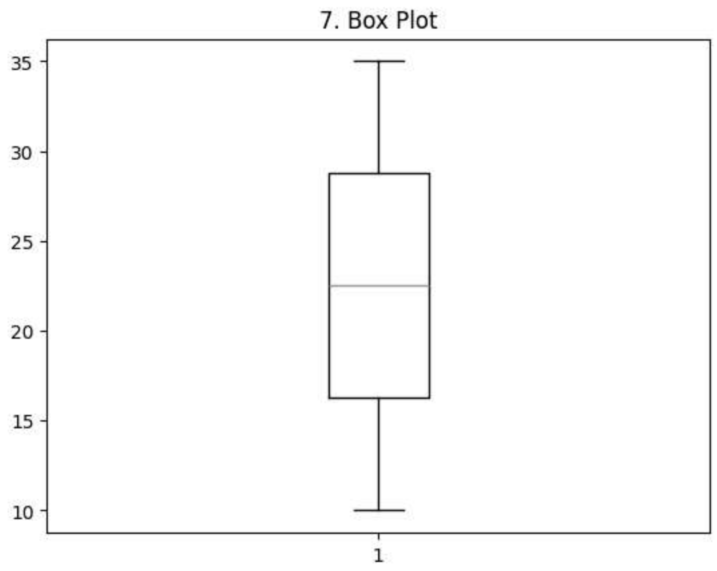

3. Bar Chart
10
8
6
4
2
0
1
2
3
4
5

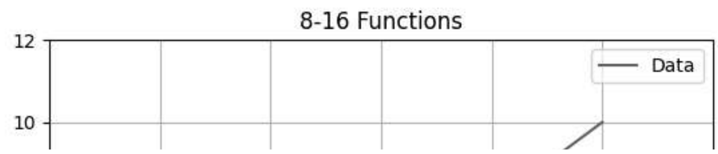

4. Horizontal Bar Chart
5
4
3

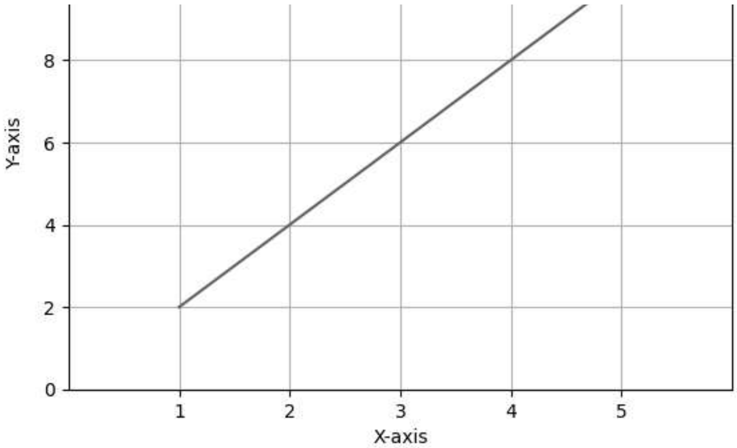

2
1
0
2
4
6
8
10

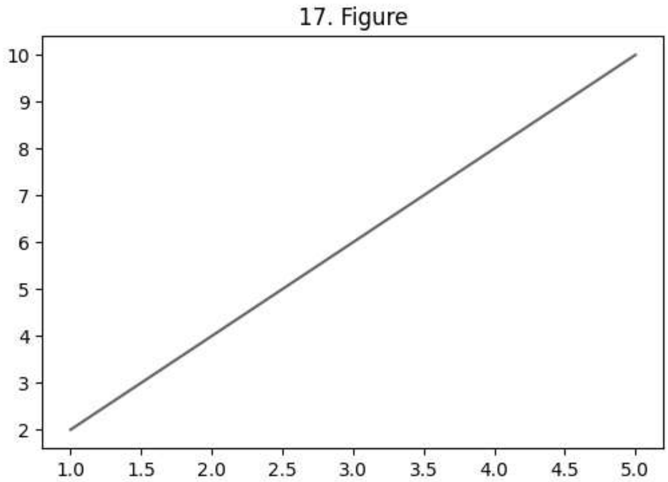

5. Histogram
3.0
2.5
2.0
1.5
1.0
0.5
0.0
1.0
1.5
2.0
2.5
3.0
3.5
4.0
4.5
5.0

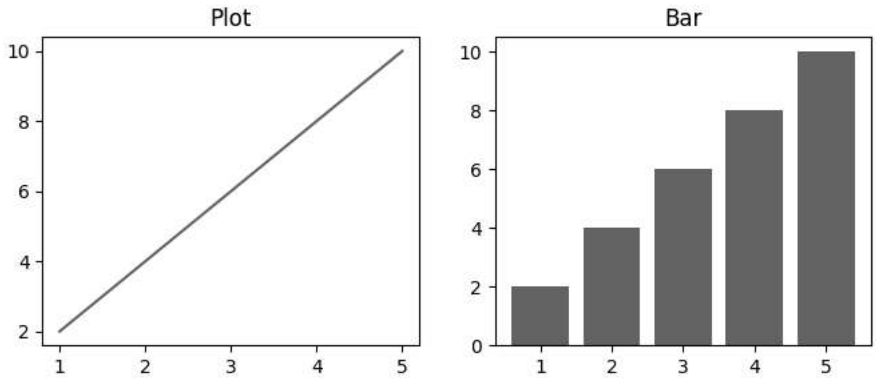

6. Pie Chart
A
30.0%
B
40.0%
30.0%
C

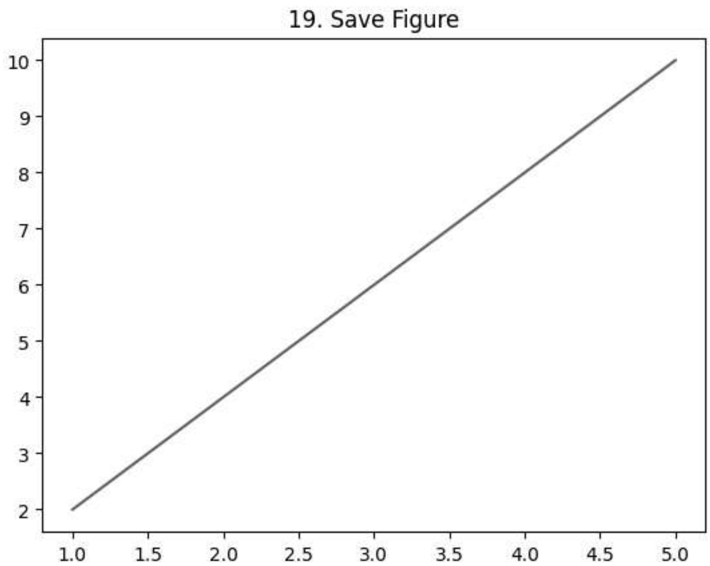

7. Box Plot
35
30
25
20
15
10
1

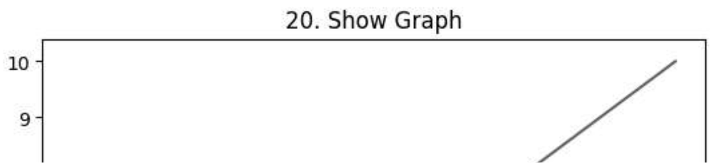

8-16 Functions
12
10
Data


Y-axis
X-axis


17. Figure


Plot
Bar


19. Save Figure


20. Show Graph


8
7
6